

2017-Ph-H1-Q7

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

An object falls from rest. What happens to its kinetic energy and total mechanical energy?

Key Concept:

- **Kinetic energy** increases (as object speeds up)
- **Gravitational potential energy** decreases
- **Total mechanical energy** (KE + PE) stays **constant** if no losses

 **Final Answer: A — Kinetic energy increases, Total energy constant**

Revision Tips:

- In free fall:

KE increases as GPE decreases

- Total energy is conserved if **no air resistance**